Physics I

Code : EBA1110

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Course Content

- Lecture 1: Units and Vectors.
- Lecture 2: Newton Laws of motion &Gravitational Force, weight and mass
- Lecture 3: Work, Kinetic and Potential Energy, Work Energy theorem& Conservation law of Energy

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- Lecture 4: Oscillatory motion , simple and Harmonic Motion, Introduction to mechanical waves properties , and Travelling wave
- Lecture 5: Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation
- Lecture 6: Superposition of waves.

Course Content

- Lecture 7: capacitors.
- Lecture 8: Calculation of the Electric Current, Resistance, Ohm's law, Power Calculation.
- Lecture 9: Electromotive force , Internal Resistance , Resistance in Series and Parallel, Kirchhoff law and RC Circuit
- Lecture 10: Magnetic Force
- Lecture 11: Biot Savart Law, Ampere Law and their applications. Magnetic force for two parallel conductors.Explanation of Faraday Law of Induction

Lecture 6

Superposition and Standing Waves



Waves in Interference

Many interesting wave phenomena in nature cannot be described by a single traveling wave. Instead, one must analyze these phenomena in terms of a combination of traveling waves.
waves have a remarkable difference from particles in that waves can be combined at the same location in space. To analyze such wave combinations, we make use of the superposition principle

Superposition principle

If two or more traveling waves are moving through a medium, the resultant value of the wave function at any point is the algebraic sum of the values of the wave functions of the individual waves.

- Waves that obey this principle are called linear waves. In the case of mechanical waves, linear waves are generally characterized by having amplitudes much smaller than their wavelengths.
- Waves that violate the superposition principle are called nonlinear waves and are often characterized by large amplitudes.

interference.

The combination of separate waves in the same region of space to produce a resultant wave

interference

constructive interference

destructive interference

For the two pulses shown in Figure 18.1, the displacement of the elements of the medium is in the positive y direction for both pulses, and the resultant pulse (created when the individual pulses overlap) exhibits an amplitude greater than that of either individual pulse. Because the displacements caused by the two pulses are in the same direction we refer to their superposition as constructive interference.

consider two pulses traveling in opposite directions on a taut string where one pulse is inverted relative to the other as illustrated in Figure 18.2. When these pulses begin to overlap, the resultant pulse is given by y1 1 y2, but the values of the function y2 are negative. Again, the two pulses pass through each other; because the displacements caused by the two pulses are in opposite directions, however, we refer to their superposition as destructive interference.

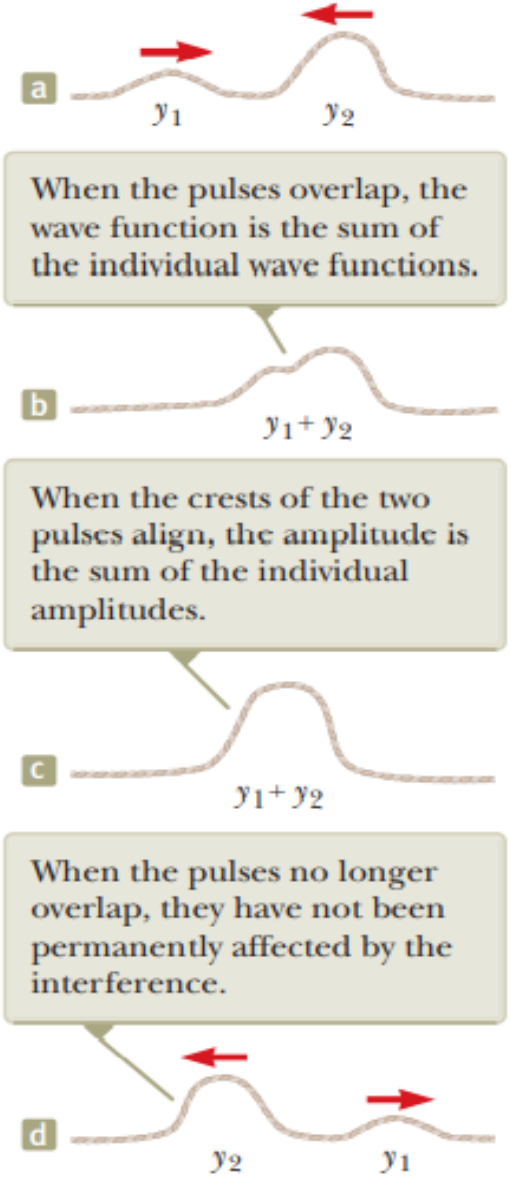


Figure 18.1 Constructive interference. Two positive pulses travel on a stretched string in opposite directions and overlap.

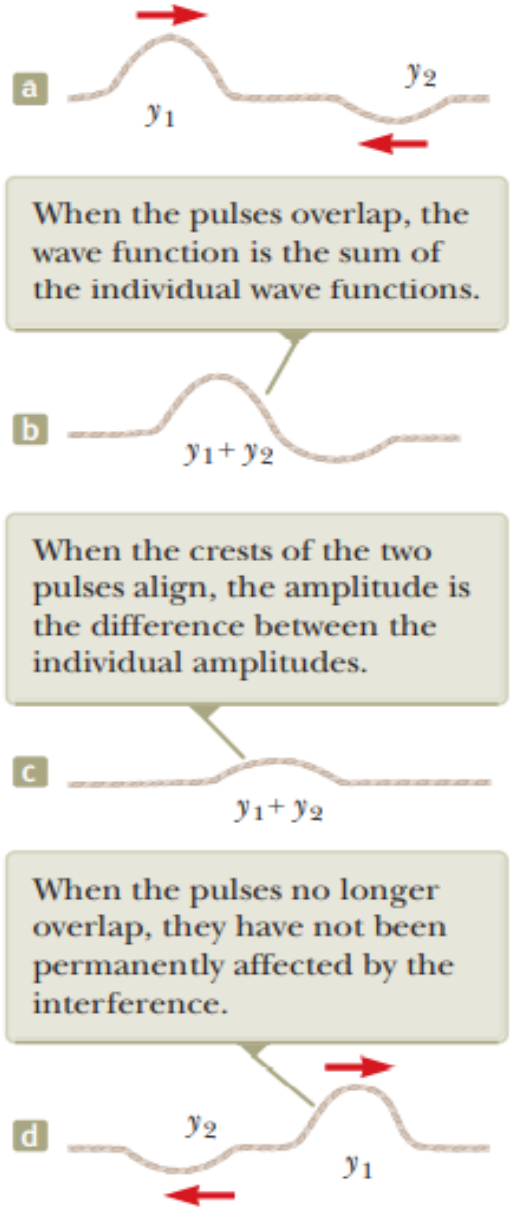


Figure 18.2 Destructive interference. Two pulses, one positive and one negative, travel on a stretched string in opposite directions and overlap.

Superposition of Sinusoidal Waves

If two waves are traveling to the right and have the same frequency, wavelength, and amplitude but differ in phase, we can express their individual wave functions as

$$y_1 = A \sin (kx - \omega t) \quad y_2 = A \sin (kx - \omega t + \phi)$$

where, as usual, $k = 2\pi/\lambda$, $\omega = 2\pi f$, and ϕ is the phase constant as discussed in Section 16.2. Hence, the resultant wave function y is

$$y = y_1 + y_2 = A [\sin (kx - \omega t) + \sin (kx - \omega t + \phi)]$$

By using the trigonometric identity

$$\sin a + \sin b = 2 \cos \left(\frac{a - b}{2} \right) \sin \left(\frac{a + b}{2} \right)$$

then

$$y = 2A \cos \left(\frac{\phi}{2} \right) \sin \left(kx - \omega t + \frac{\phi}{2} \right)$$

This result has several important features.

- The resultant wave function y also is sinusoidal and has the same frequency and wavelength as the individual waves because the sine function incorporates the same values of k and ω that appear in the original wave functions.
- The amplitude of the resultant wave is 2A cos (Φ/2), and its phase constant is Φ/2.
- If the phase constant Φ of the original wave equals 0, then cos (Φ/2) = cos 0 = 1 and the amplitude of the resultant wave is 2A, twice the amplitude of either individual wave.
- In this case, the crests of the two waves are at the same locations in space and the waves are said to be everywhere in phase and therefore interfere constructively.
- The individual waves y1 and y2 combine to form the red-brown curve y of amplitude 2A shown in Figure 18.3a. Because the individual waves are in phase, they are indistinguishable in Figure 18.3a, where they appear as a single blue curve.
- In general, constructive interference occurs when cos (Φ/2) = ∓1. That is true, for example, when Φ = 0, 2π, 4π, . . . rad, that is, when Φ is an even multiple of π.

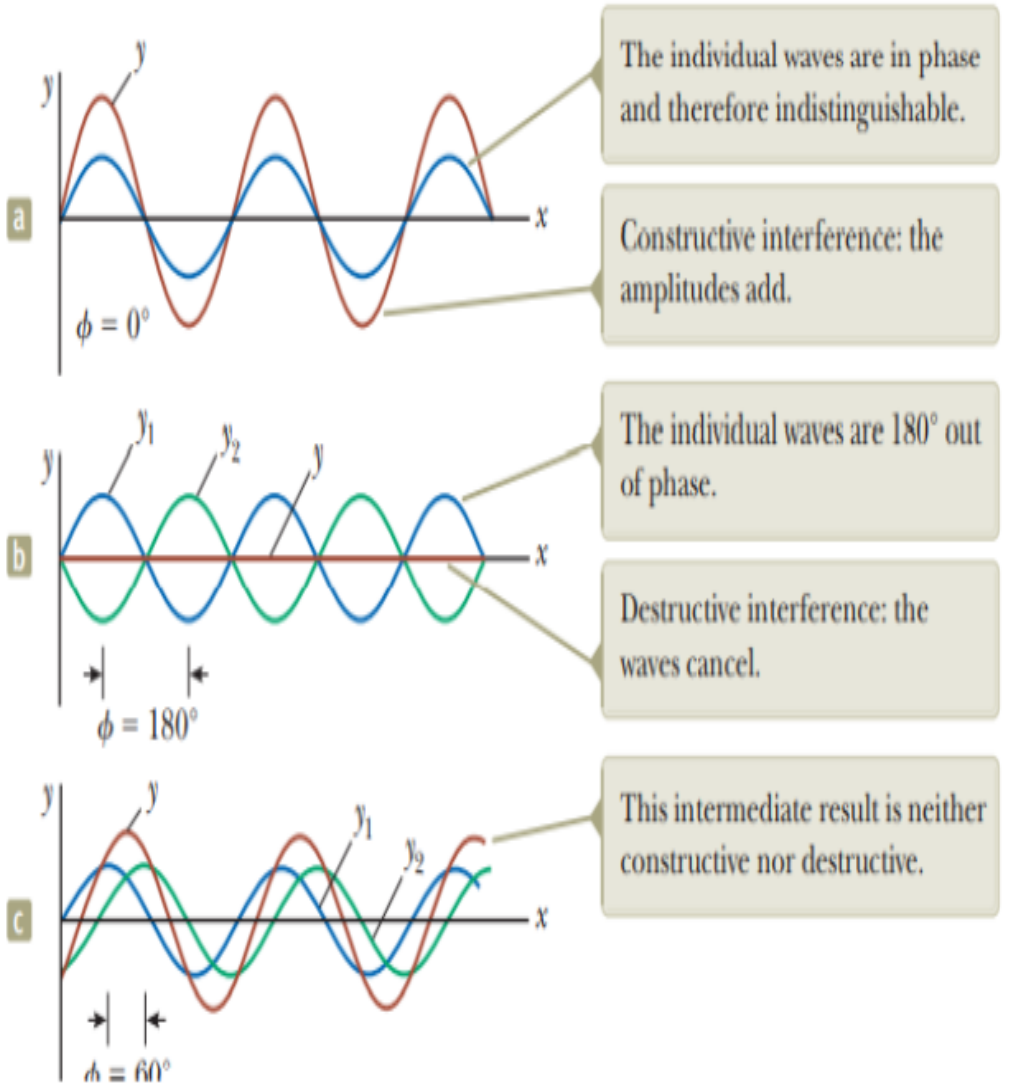


Figure 18.3 The superposition of two identical waves y_1 and y_2 (blue and green, respectively) to yield a resultant wave (red-brown).

When Φ is equal to π rad or to any odd multiple of π, then cos (Φ/2) = cos (π/2) =0 and the crests of one wave occur at the same positions as the troughs of the second wave (Fig. 18.3b). Therefore, as a consequence of destructive interference, the resultant wave has zero amplitude everywhere as shown by the straight red-brown line in Figure 18.3b.

Finally, when the phase constant has an arbitrary value other than 0 or an integer multiple of π rad (Fig. 18.3c), the resultant wave has an amplitude whose value is somewhere between 0 and 2A.

In the more general case in which the waves have the same wavelength but different amplitudes, the results are similar with the following exceptions. In the inphase case, the amplitude of the resultant wave is not twice that of a single wave, but rather is the sum of the amplitudes of the two waves. When the waves are π rad out of phase, they do not completely cancel as in Figure 18.3b. The result is a wave whose amplitude is the difference in the amplitudes of the individual waves.

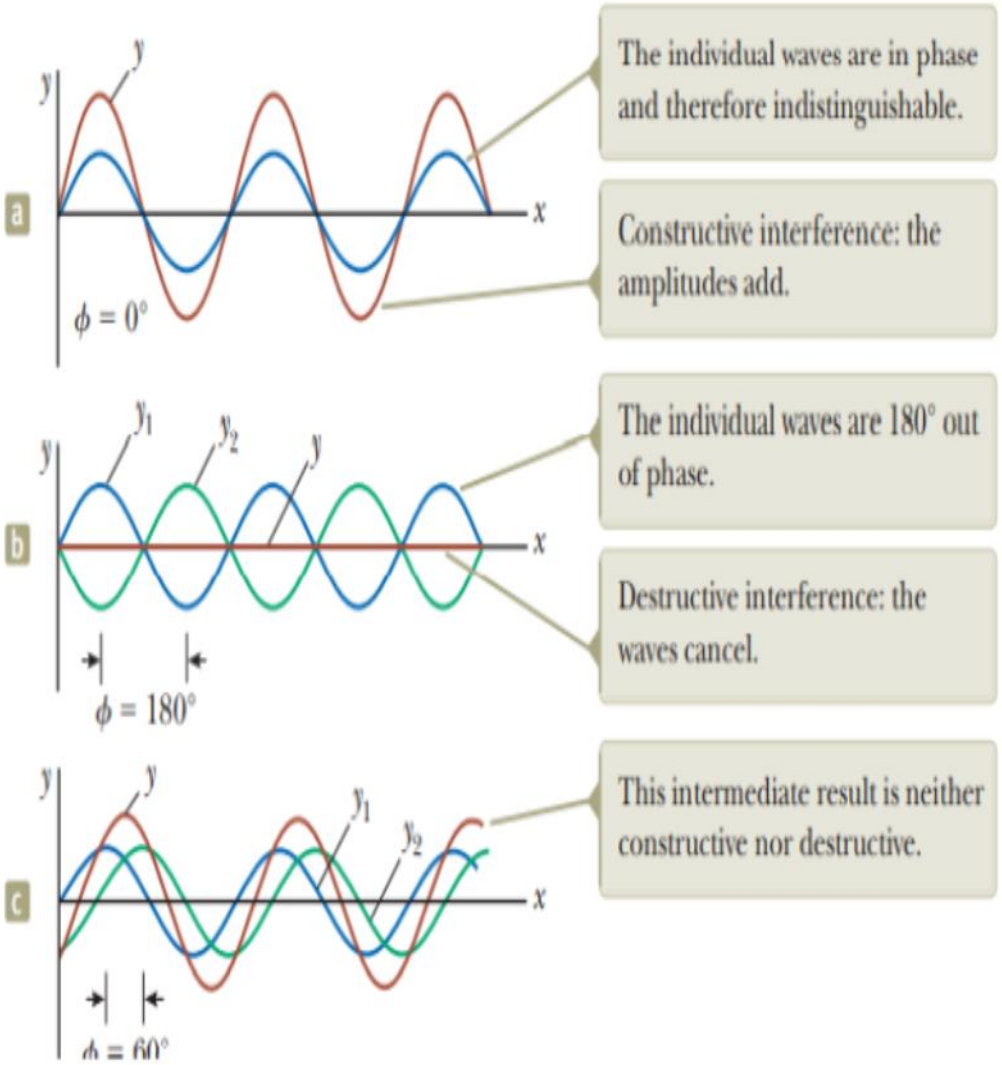


Figure 18.3 The superposition of two identical waves y_1 and y_2 (blue and green, respectively) to yield a resultant wave (red-brown).

Interference of Sound Waves

One simple device for demonstrating interference of sound waves is illustrated in Figure 18.4.
Sound from a loudspeaker *S* is sent into a tube at point *P*, where there is a T-shaped junction.
Half the sound energy travels in one direction, and half travels in the opposite direction. Therefore, the sound waves that reach the receiver R can travel along either of the two paths.
The distance along any path from speaker to receiver is called the path length r. The lower path length r_1 is fixed, but the upper path length r_2 can be varied by sliding the U-shaped tube, which is similar to that on a slide trombone.
When the difference in the path lengths $\Delta r = |r_2 - r_1|$ is either zero or some integer multiple of the wavelength λ(that is, $\Delta r = n\lambda$, where $n = 0, 1, 2, 3, \ldots$), the two waves reaching the receiver at any instant are in phase and interfere constructively as shown in Figure 18.3a. For this case, a maximum in the sound intensity is detected at the receiver.
If the path length r_2 is adjusted such that the path difference $\Delta r = \lambda/2, 3\lambda/2, \ldots, n\lambda/2$ (for n odd), the two waves are exactly π rad, or 180°, out of phase at the receiver and hence cancel each other. In this case of destructive interference, no sound is detected at the receiver.
This simple experiment demonstrates that a phase difference may arise between two waves generated by the same source when they travel along paths of unequal lengths.

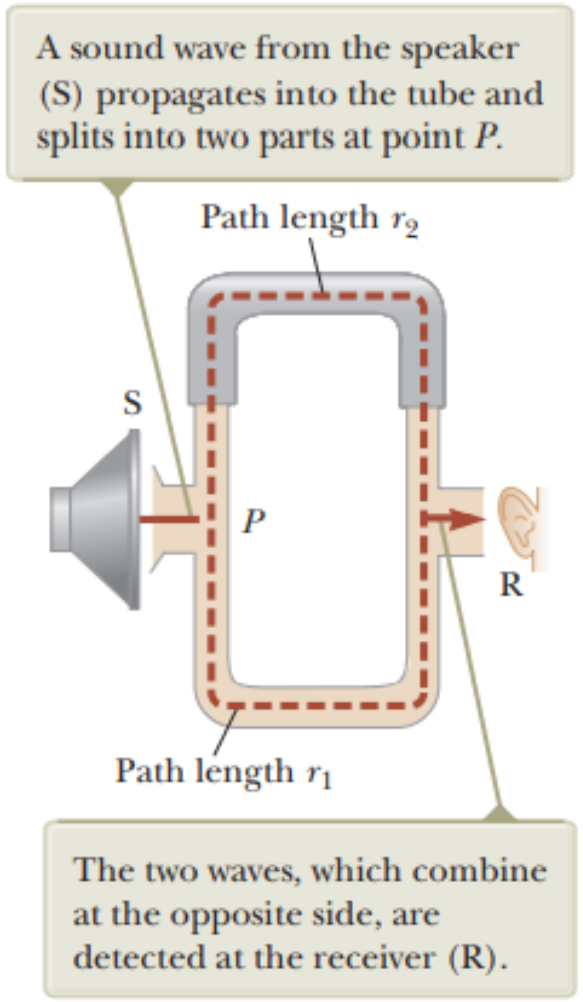


Figure 18.4 An acoustical system for demonstrating interference of sound waves. The upper path length r_2 can be varied by sliding the upper section.

Example 18.1 Two Speakers Driven by the Same Source AM

Two identical loudspeakers placed 3.00 m apart are driven by the same oscillator (Fig. 18.5). A listener is originally at point *O*, located 8.00 m from the center of the line connecting the two speakers. The listener then moves to point *P*, which is a perpendicular distance 0.350 m from *O*, and she experiences the *first minimum* in sound intensity. What is the frequency of the oscillator?

SOLUTION

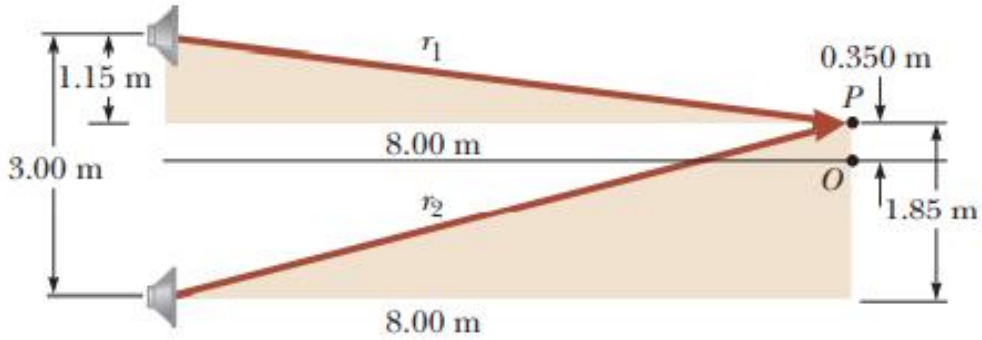


Figure 18.5 (Example 18.1) Two identical loudspeakers emit sound waves to a listener at *P*.

Analyze Figure 18.5 shows the physical arrangement of the speakers, along with two shaded right triangles that can be drawn on the basis of the lengths described in the problem. The first minimum occurs when the two waves reaching the listener at point *P* are 180° out of phase, in other words, when their path difference Δr equals $\lambda/2$.

From the shaded triangles, find the path lengths from the speakers to the listener:

$$r_1 = \sqrt{(8.00\ \text{m})^2 + (1.15\ \text{m})^2} = 8.08\ \text{m}$$

$$r_2 = \sqrt{(8.00\ \text{m})^2 + (1.85\ \text{m})^2} = 8.21\ \text{m}$$

Hence, the path difference is $r_2 - r_1 = 0.13$ m. Because this path difference must equal $\lambda/2$ for the first minimum, $\lambda = 0.26$ m.

To obtain the oscillator frequency, use Equation 16.12, $v = \lambda f$, where v is the speed of sound in air, 343 m/s:

$$f = \frac{v}{\lambda} = \frac{343\ \text{m/s}}{0.26\ \text{m}} = 1.3\ \text{kHz}$$

Standing Waves

is an oscillation pattern with a stationary outline that results from the superposition of two identical waves traveling in opposite directions.

such as the one on a string shown in Figure 18.7

We can analyze such a situation by considering wave functions for two transverse sinusoidal waves having the same amplitude, frequency, and wavelength but traveling in opposite directions in the same medium:

$$y_1 = A \sin (kx - \omega t) \quad y_2 = A \sin (kx + \omega t)$$

where y_1 represents a wave traveling in the positive x direction and y_2 represents one traveling in the negative x direction. Adding these two functions gives the resultant wave function y:

$$y = y_1 + y_2 = A \sin (kx - \omega t) + A \sin (kx + \omega t)$$

When we use the trigonometric identity $\sin (a \pm b) = \sin a \cos b \pm \cos a \sin b$, this expression reduces to

$$y = (2A \sin kx) \cos \omega t$$

Equation 18.1 represents the wave function of a **standing wave.** A standing wave

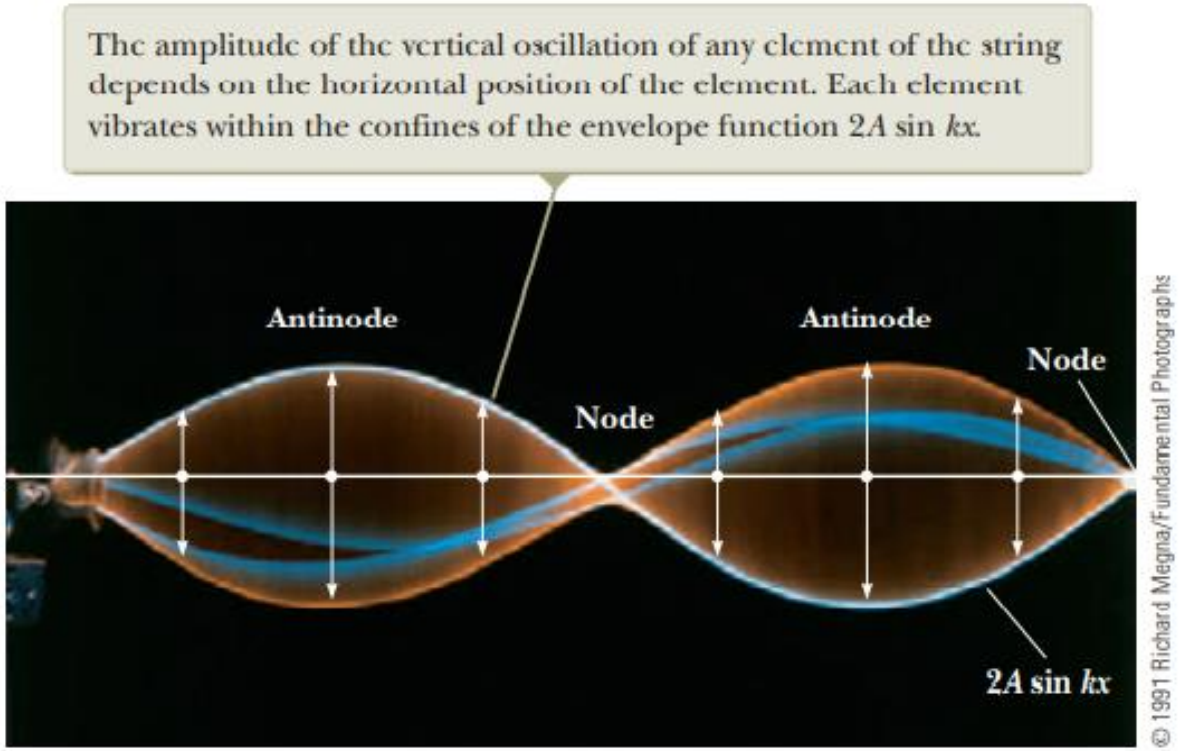


Figure 18.7 Multiflash photograph of a standing wave on a string. The time behavior of the vertical displacement from equilibrium of an individual element of the string is given by cos ωt. That is, each element vibrates at an angular frequency ω.

Notice that

- Equation 18.1 does not contain a function of kx -ωt. Therefore, it is not an expression for a single traveling wave.
- When you observe a standing wave, there is no sense of motion in the direction of propagation of either original wave.
- Comparing Equation 18.1 with Equation 15.6, we see that it describes a special kind
- of simple harmonic motion.
- Every element of the medium oscillates in simple harmonic motion with the same angular frequency ω (according to the cos ωt factor in the equation).
- The amplitude of the simple harmonic motion of a given element (given by the factor 2A sin kx, the coefficient of the cosine function) depends on the location x of the element in the medium, however.

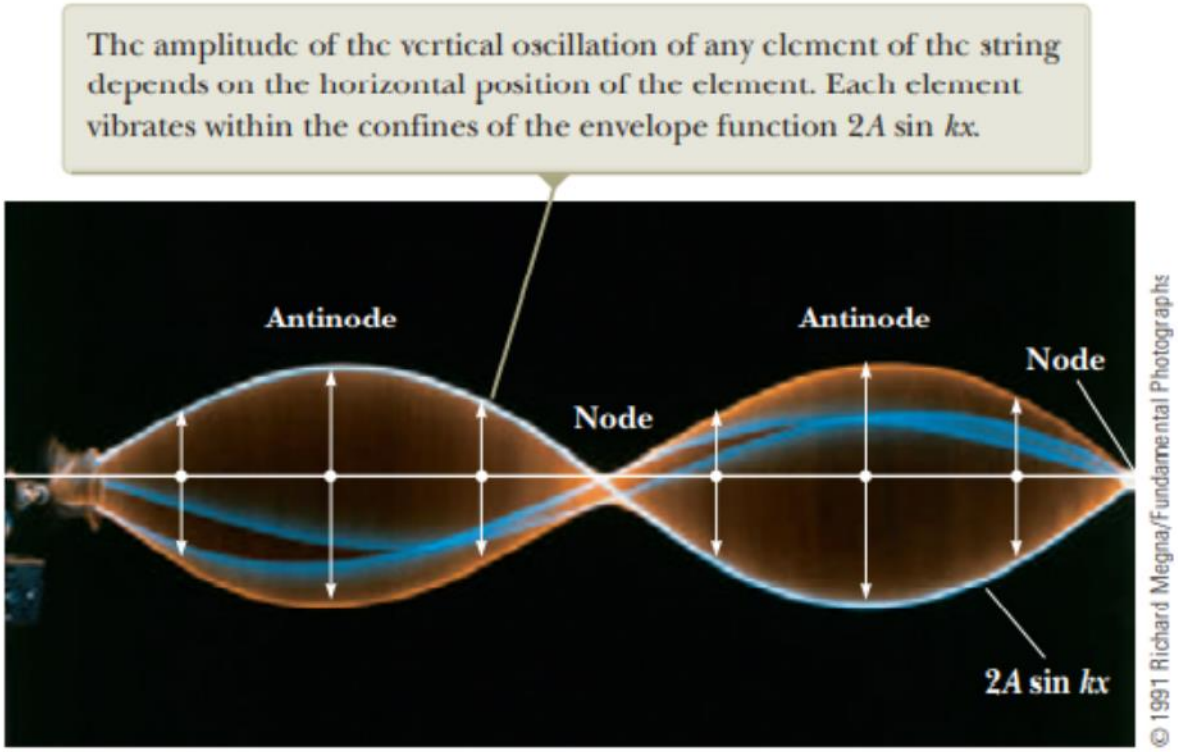


Figure 18.7 Multiflash photograph of a standing wave on a string. The time behavior of the vertical displacement from equilibrium of an individual element of the string is given by cos ωt. That is, each element vibrates at an angular frequency ω.

Equation 18.1 shows that the amplitude of the simple harmonic motion of an element of the medium has a minimum value of zero when x satisfies the condition $\sin kx = 0$, that is, when

$$kx = 0, \pi, 2\pi, 3\pi, \ldots$$

Because $k = 2\pi/\lambda$, these values for kx give

$$x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, \ldots = \frac{n\lambda}{2} \qquad n = 0, 1, 2, 3, \ldots \tag{18.2}$$

These points of zero amplitude are called **nodes.**

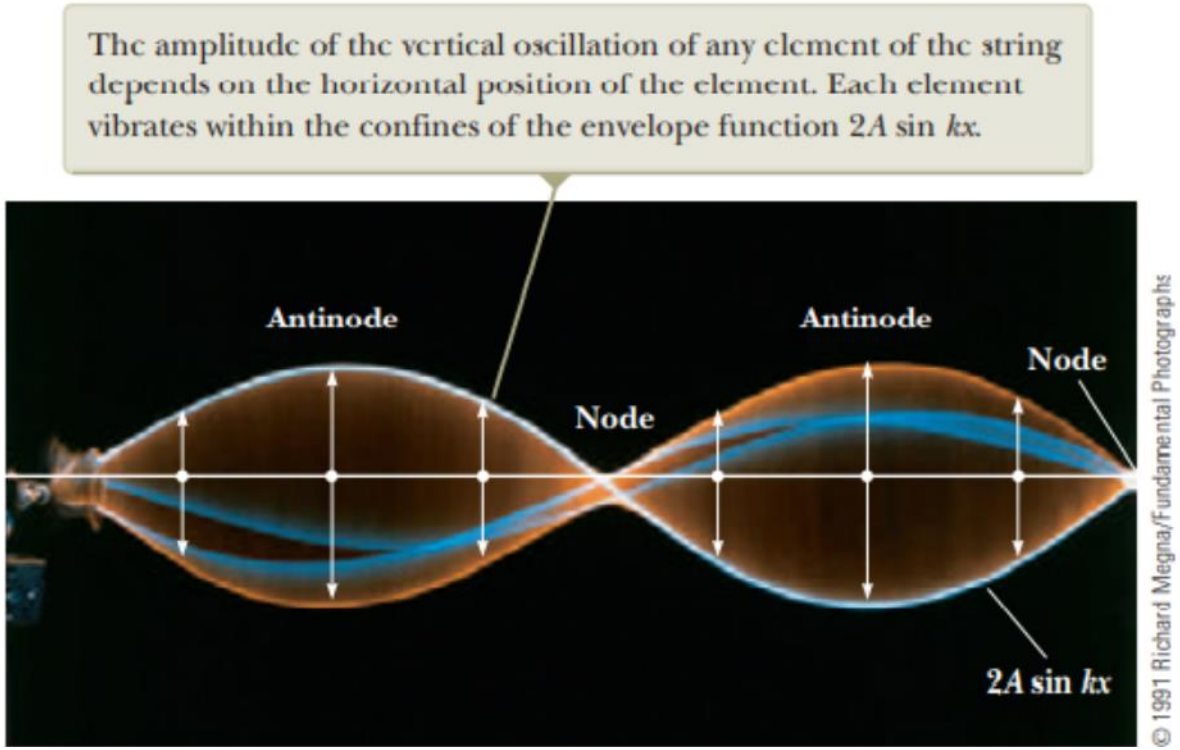


Figure 18.7 Multiflash photograph of a standing wave on a string. The time behavior of the vertical displacement from equilibrium of an individual element of the string is given by cos ωt. That is, each element vibrates at an angular frequency ω.

The element of the medium with the *greatest* possible displacement from equilibrium has an amplitude of $2A$, which we define as the amplitude of the standing wave.

The positions in the medium at which this maximum displacement occurs are called **antinodes.**

The antinodes are located at positions for which the coordinate x satisfies the condition $\sin kx = \pm 1$, that is, when

$$kx = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$$

Therefore, the positions of the antinodes are given by

$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots = \frac{n\lambda}{4} \quad n = 1, 3, 5, \ldots \tag{18.3}$$

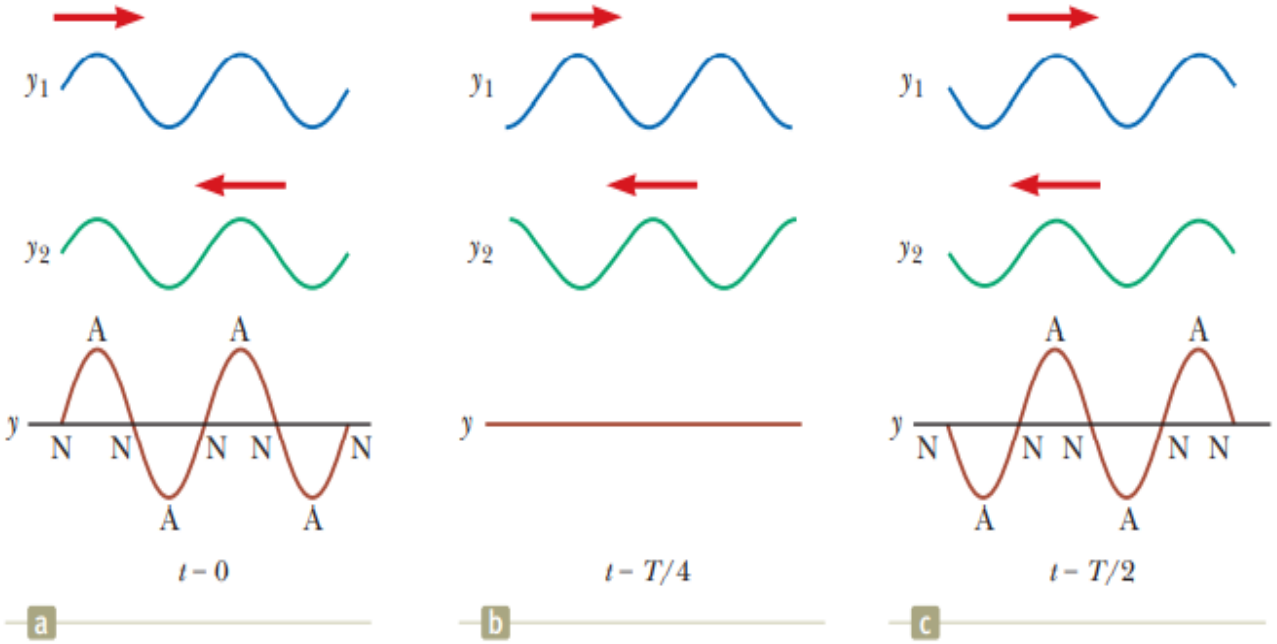


Figure 18.8 Standing-wave patterns produced at various times by two waves of equal amplitude traveling in opposite directions. For the resultant wave y, the nodes (N) are points of zero displacement and the antinodes (A) are points of maximum displacement.

The distance between adjacent antinodes is equal to $\lambda/2$.
The distance between adjacent nodes is equal to $\lambda/2$.
The distance between a node and an adjacent antinode is $\lambda/4$.

Waves Under Boundary Conditions

1-standing waves set up in the string

Consider a string of length L fixed at both ends as shown in Figure 18.9. Waves can travel in both directions on the string. Therefore, standing waves can be set up in the string by a continuous superposition of waves incident on and reflected from the ends. Notice that there is a boundary condition for the waves on the string: because the ends of the string are fixed, they must necessarily have zero displacement and are therefore nodes by definition. The condition that both ends of the string must be nodes fixes the wavelength of the standing wave on the string according to Equation 18.2, which, in turn, determines the frequency of the wave. The boundary condition results in the string having a number of discrete natural patterns of oscillation, called normal modes, each of which has a characteristic frequency that is easily calculated. This situation in which only certain frequencies of oscillation are allowed is called quantization.

Notice there is no quantization without boundary conditions.

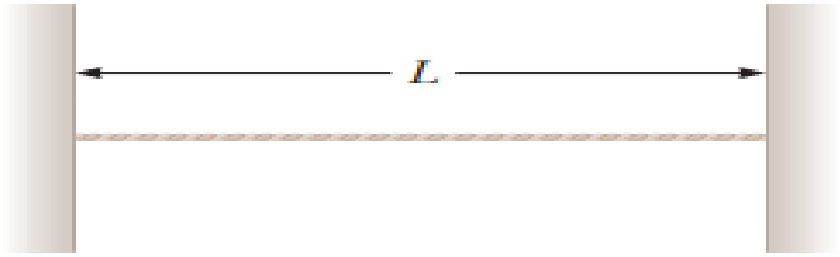


Figure 18.9 A string of length L fixed at both ends.

The normal modes of oscillation for the string in Figure 18.9 can be described by imposing the boundary conditions that the ends be nodes and that the nodes be separated by one-half of a wavelength with antinodes halfway between the nodes.
The first normal mode that is consistent with these requirements, shown in Figure 18.10a ,has nodes at its ends and one antinode in the middle. This normal mode is the longest-wavelength mode that is consistent with our boundary conditions. The first normal mode occurs when the wavelength λ1 is equal to twice the length of the string, or λ1 = 2L. The section of a standing wave from one node to the next node is called a loop. In the first normal mode, the string is vibrating in one loop.

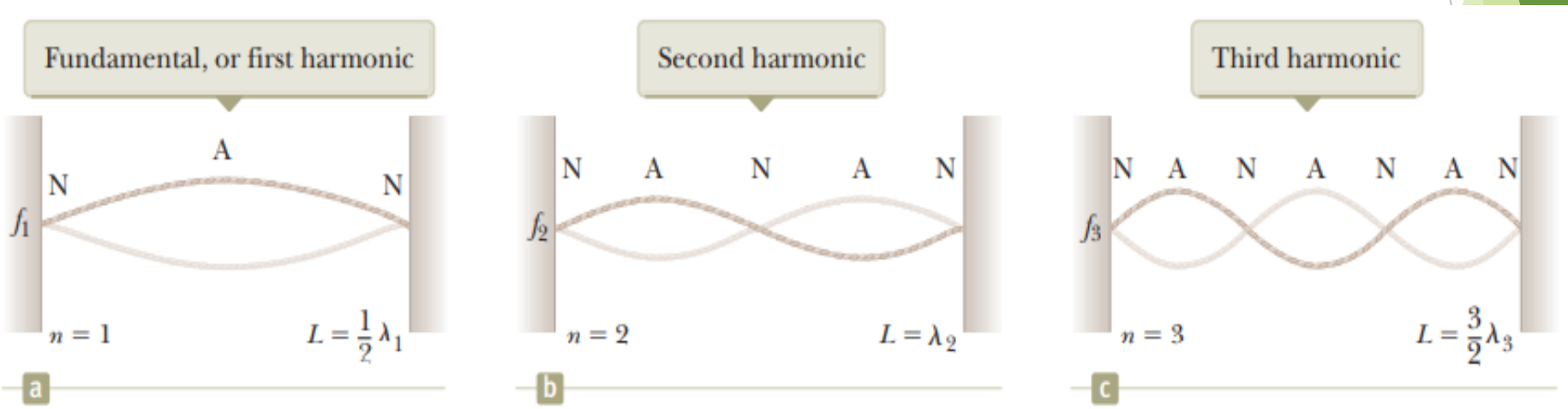


Figure 18.10 The normal modes of vibration of the string in Figure 18.9 form a harmonic series. The string vibrates between the extremes shown.

In the second normal mode (see Fig. 18.10b), the string vibrates in two loops. When the left half of the string is moving upward, the right half is moving downward. In this case, the wavelength λ_2 is equal to the length of the string, as expressed by $\lambda_2 = L$.

The third normal mode (see Fig. 18.10c) corresponds to the case in which $\lambda_3 = 2L/3$, and the string vibrates in three loops.

In general, the wavelengths of the various normal modes for a string of length L fixed at both ends

$$\lambda_n = \frac{2L}{n} \quad n = 1, 2, 3, \ldots$$

Using Equation 18.4, we find that the natural frequencies f_n of the normal modes are

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2L} \quad n = 1, 2, 3, \ldots \tag{18.5}$$

Because $v = \sqrt{T/\mu}$ for waves on a string, where T is the tension in the string and μ is its linear mass density, we can also express the natural frequencies of a taut string as

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}} \quad n = 1, 2, 3, \ldots \tag{18.6}$$

The lowest frequency f_1, which corresponds to $n = 1$, is called either the **fundamental** or the **fundamental frequency** and is given by

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}} \tag{18.7}$$